

- State and explain the types of disaster mitigation measures and also discuss the role of international relief organizations in disaster response and recovery.

Disaster mitigation refers to the lessening or limitation of the adverse impacts of hazards and related disasters.

Types of Disaster Mitigation Measures

- Structural Mitigation Measures: Any physical construction to reduce or avoid possible impacts of hazards, which includes engineering measures and construction of hazard resistant and protective structures and infrastructure.
- Non-structural Mitigation Measures: This refers to policies, awareness, knowledge development, public commitment, and methods and operating practices, including participatory mechanisms and the provision of information, which can reduce risk and related impacts.
- Structural and non-structural disaster mitigation measures are mainly carried out by human beings.
- However, nature through ecosystem functions also provides several mitigation measures as shown in next slide.

Hazard and disaster mitigation functions of ecosystems

Ecosystem	Hazard Mitigation
Mountain forests, vegetation on hillsides	Vegetation cover and root structures protect against erosion and increase slope stability by binding soil together, preventing landslides. Catchment forests, especially primary forests, reduce risk of floods by increasing infiltration of rainfall and delaying peak floodwater flows, except when soils are fully Saturated. Forests on watersheds are important for water recharge and purification, drought mitigation and safeguarding drinking water supply.

Hazard and disaster mitigation functions of ecosystems

Ecosystem	Hazard Mitigation
Wetlands, floodplains	Wetlands and floodplains control floods in coastal areas. Marshes, lakes and floodplains release wet season flows slowly during drought periods.
Coastal ecosystems (mangroves, saltmarshes, coral reefs, sand dunes)	Coastal wetlands, tidal flats, deltas and estuaries reduce the height and speed of storm surges and tidal waves. Coastal ecosystems protect against storm surges, flooding and other coastal hazards – combined protection by coral reefs, seagrass beds and sand dunes/ coastal wetlands/coastal forests is particularly effective.

Hazard and disaster mitigation functions of ecosystems

Ecosystem	Hazard Mitigation
Drylands	Natural vegetation management and restoration in drylands contributes to ameliorate the effects of drought and control desertification, as trees, grasses and shrubs conserve soil and retain moisture. Prescribed burning and creation of physical firebreaks in dry landscapes reduce fuel loads and the risk of unwanted large-scale fires.

International relief organizations play a crucial role in disaster response and recovery by providing immediate assistance and long-term support. During crises, these organizations mobilize resources, including food, medical supplies, and personnel, to assist affected populations. They facilitate coordination among various stakeholders, including governments and local NGOs, to ensure effective and efficient response efforts. In the recovery phase, international organizations often assist in rebuilding infrastructure, restoring services, and supporting community resilience through capacity-building initiatives. Their expertise and resources are vital in both immediate

relief efforts and long-term recovery strategies, helping communities to not only recover but also to better prepare for future disasters.

- Identify the factors that determine the nature of disaster response and explain.
 - Factors that Determine the Nature of Disaster Response
 - The type of disaster: Disaster manifests in many forms, its onset may provide long warning, short warning, or no warning at all. It would influence the effectiveness of activating preparedness plans, mobilisation, and application or response effort.
 - The ability to take pre-impact actions: Responses to disaster are operationalized in three main phases namely the “pre-during and post-disaster” situation. Disaster early warning systems may provide timely warnings for anticipating impending disaster. Pre-impact responses may be carried out if time and conditions are favourable.
 - The severity and magnitude of disaster: The severity and size of the problem determines the response required. Particular effects could be seen in the ability of responses to cope with the problem.
 - The capability of sustained operations: This is influenced by factors such as resource capability, management capability, community self-reliance, and availability of international assistance. It is important that these issues are clearly addressed in preparedness planning and response action plans.
 - Identification of likely response requirements: It is to be identified in the preparedness planning stage. The characteristics of disaster events might result in some or all of the following:
 - Many urgent and significant tasks involving injuries, deaths, or property loss/damage.
 - Large numbers of personnel and agencies required.
 - High levels of public interest and/or controversy.
 - Many examples of information mishandling.
 - Potential for some key tasks to be overlooked, under-resourced, over-resourced.
 - Potential for some of the available key resources to not be used.
 - Factors that Determine the Nature of Disaster Response
 - Requirement for Effective Response
 - Information: An early warning system provides vital information for effective response operation. It must be robust to transmit warnings as early as practicable. Information gained from these systems could help in the planning and decision-making as well as inform the general public.
- Explain the different disaster response actions

Disaster responses include actions that embrace the following:

- > Search and rescue
- > First aid and emergency medical care
- > Evacuation
- > Evacuation centre management
- > Development of Standard Operation Procedure (SOPs)
- > Immediate repair of community facilities and services
- > Relief delivery
- > Coordination and Communication
- > Psycho-social counselling and stress debriefing
- > Medical services

- Search and rescue

- This activity is usually conducted by well-trained volunteers in finding disaster victims, that is, lost, sick or injured persons in either a remote or difficult to access areas such as water bodies, desert, forest or probably in the course of mass population movement.

- They are often directed at, locating endangered persons at an emergency incident, removing those persons from danger, treating the injured, and providing for transport to an appropriate health care facility.

- Basic Key Steps of Search and Rescue

1. Size up - involves assessing the situation and determining what one is going to do and if yes, then how.

The decision whether to attempt a rescue should be based on:

- a. The risks involved; and
- b. Achievement of the overall goal of doing the greatest good for the greatest number.

2. Search - involves locating victims and documenting their location.

3. Rescue - involves the procedures and methods of extricating and moving victims to safety.

- First aid and emergency medical care

- > First aid is the provision of initial care for an illness or injury.

- > It is usually performed by non-expert, but trained personnel to a sick or injured person until definitive medical treatment can be accessed.

- > Emergency medical care is immediate paramedic attention to severe wounds and the rapid transportation of the ill or injured to a health facility.

- Evacuation

Evacuation is an organised movement of people from an area at risk to a safer place.

Types of Evacuation

- Precautionary evacuation before disaster

- Protective evacuation after disaster
- Evacuations for reconstruction purposes

- Evacuation

Services provided during evacuations include:

- Registration
- Assistance with financial and legal queries
- Water, food, clothing
- Rest areas
- Blankets and personal items
- Interpreter services
- Assistance in contacting family/friends
- Services for animals
- Emergency financial assistance
- First aid, medical and health
- Information

- Standard Operating Procedures (SOPs)

SOPs are the set of standard procedures that

“operationalize” the disaster response and/or contingency/ plans. The SOPs set out what should be done, how it should be done, who is responsible for implementing what, and specifies available resources.

• SOPs take cognisance of four stages of preparation and procedures:

- > during normal times
- > alert/warning and
- > during disaster
- > rehabilitation

❖ During Normal Times

The state institution mandated to respond to disaster ought to:

- > Formulate and distribute disaster preparedness plans, and conduct drills in all areas;
- > Produce maps of Wards/Village Tracts showing areas most vulnerable to storms, floods and other natural disasters;
- > Make a list of vehicles and motor boats that can be used for emergency work;
- > Compile a list of departments, non-governmental organisations (NGOs), and members of People’s Strength that will take part in relief operations in the predisaster, disaster and post-disaster periods and designate representatives for contact;

❖ Alert/Warning Stage

- Emphasize the dissemination of news obtained through early warning systems to the community;
- Assign duties to administrative bodies and NGOs to fly warning flags as part of the disaster preparedness programme in the vulnerable areas of the Ward/Village Tract;
- Alert and mobilise members of the Security services, Auxiliary Fire Brigade, communication agencies, the Red Cross, Youth, members of People's Strength and NGOs
- Make the necessary arrangements to evacuate the public to safe locations (shelters) in a timely manner;
- Increase security sentries as required;
- Ensure that all levels of supervisors have all teams ready for assigned duties; and
- Keep the office operational 24 hours a day in the emergency period.

❖ During Disaster Stage

- Alert the community in areas the natural disaster is likely to strike;
- Safeguard the road and water transport routes, keep relief and medical teams at the ready and arrange transport to affected areas at short notice;
- Evacuate the community from vulnerable areas to safe locations or designated shelters as quickly as possible;
- Operate relief camps and supervisory centers at designated shelters as quickly as possible;
- Ensure that administrative personnel and NGOs in areas vulnerable to storms give disaster warnings door to door as a matter of urgency
- Keep available relief and aid supplies at the ready to launch relief operations quickly and effectively;
- Evacuate the public remaining in the area to designated safe locations;
- Make arrangements to evacuate movable property including cattle to designated locations; and
- Ensure the well-disciplined implementation of orders received from the coordinating agencies and sub-committees with the help of members of the Security services, Fire Brigade, Red Cross Youth members, and members of People's Strength, social organisations and NGOs.

❖ Rehabilitation Stage

The SOPs for the rehabilitation stage are:

- Conduct field inspections in affected areas as soon as possible and provide the necessary assistance and support;
- Submit immediate preliminary reports with population figures, death and injury figures of cattle and animals, data on socio-economic losses, and carry out further systematic data collection;
- Make arrangements to provide health care and social protection to disaster victims;
- Clear collapsed buildings and trees as quickly as possible;
- Prioritise the restoration of transportation, electricity and water supply and telephone and telegraph services as soon as possible

- Make arrangements as quickly as possible to reclaim contaminated wells and ponds for access to clean water and dig new wells for drinking water;
- Make arrangements to bury/cremate the remains of disaster casualties and animal carcasses;
- Manage and systematically utilise disaster funds and supplies, as well as cash and supplies donated by well-wishers, social organisations and NGOs; and
- Support the local population for the resumption and recovery of economic activities to previous conditions.

- Relief Aid

- This relates to any provision of assistance during an emergency that is meant to attend to a person's immediate requirements for survival or recovery.
- It may include food, clothing, housing, medical care, necessary social services and security when a person is faced with circumstances beyond her or his control.
- Relief aid must be targeted at the most vulnerable first: Vulnerable children or orphans, female or child headed households, pregnant or lactating women, sick or elderly populations.

- Coordination and Communication

- Meetings - Plan to hold regular meetings with Movement partners to determine activities and roles;
- Information management - Information sharing on disaster impact, assessment and needs through input into DMIS (Disaster Management Information System);
- Communication means - Plan for continuous information and communication flow; Movement Coordination Frameworks - Plan for strategic and operational coordination; and
- Partnership agreements - Identify existing agreements and determine additional agreements needed to meet needs.

- Coordination and Communication

➤ Accurate and comprehensive information is often a requisite without which response operations would be difficult. The setting of Emergency Operations Centres (EOC) is essential for the effective management of information. EOCs ensure that information is correctly processed according to the proven cycle of:

- acquisition of information;
- information assessment;
- decision-making; and
- dissemination of decisions and information.

- Psychosocial Support

➤ Impacts on psychosocial well-being can be both short term and long term.

➤ Psychological services play a crucial role in responding to crises that involve large populations.

➤ They help in the recovery process and reduce the development of mental health problems.

➤ Psychosocial support activities include identifying and referring individuals requiring specialised support through professional mental health services.

- Public Health Services

The relevance of medical services is most felt in when there are:

- Deaths, injuries
- Loss of clean water, shelter, sanitation, routine hygiene
- Disruption of solid waste management
- Public concern for safety
- Increased pests and vectors
- Damage to health care system
- Worsening of chronic illnesses
- Toxic/hazardous exposure
- Loss of food supply
- Standing surface water

- state and explain the steps for effective communication and Identify the barriers to effective communication

Steps to Effective Communication

- Use standard terminologies when communicating-risks, disaster, coping, resilience, vulnerable, etc.
- Request and provide clarifications when needed- allow/encourage the beneficiaries to respond to issues they are not sure of.
- The communicator should also be well informed about the situation of things within the community where the information is to be disseminated.

- Ensure statements are direct and unambiguous.
- Inform appropriate individuals when the mission or the plan changes.
- Communicate all information needed by those individual or teams external to the team.
- Use non-verbal communication appropriately.
- Use proper order when communication information.

Barriers to Effective Communication

- Non-Focus on the issue at hand, not being attentive
- Avoid interruption, show interest in what is being said
- Avoid being judgemental but make provision for feedbacks
- Pay attention to non-verbal communication
- Be conscious of individual differences
- Keep stress in check but be assertive

- Explain the characteristics of effective crisis counsellors. State the advantages of crisis counselling.

Some Characteristics of Effective Crisis Counsellors

Effective crisis counsellors should possess characteristics such as:

- Self-Awareness: knows him/her self and empathise with clients without becoming personally involved or emotional when people who have gone personal experiences come to them.
- Non-judgemental: willing to listen all through to the client without casting judgement on those in crisis.
- Non-Reactive: does not react to client's outbursts or threats but be completely supportive when client shows strong emotions.
- High Tolerance: remain calm when placed in tense and stressful situations
- Specific Training: receive specific skills and techniques in crisis counselling that are quite different from normal counselling.

Strengths of Crisis Counselling

It is brief and direct

It has modest and objective goals

It is more intense than regular counselling

- It is transitional in nature

Crisis counseling offers several advantages:

1. Immediate Support: Provides quick emotional assistance to individuals in distress, helping them cope with the immediate impact of a crisis.
2. Validation of Feelings: Acknowledges and normalizes individuals' feelings of fear, anxiety, or grief, fostering a sense of understanding and acceptance.
3. Coping Strategies: Equips individuals with practical tools and strategies to manage their emotions and stress, promoting resilience.
4. Facilitates Communication: Encourages open dialogue about thoughts and feelings, which can alleviate feelings of isolation and promote healing.
5. Prevention of Long-term Issues: Early intervention can reduce the risk of developing chronic mental health problems, such as PTSD or depression.
6. Resource Connection: Links individuals to additional support services and resources, enhancing their overall recovery and well-being.
7. Empowerment: Helps individuals regain a sense of control and agency in their lives during chaotic times, fostering empowerment and hope.

- Discuss the key features and objectives of the National Disaster Management Policy in India
How does it guide the country in managing and reducing disaster risks?

Vision

- To build a safe and disaster resilient India by developing a holistic, proactive, multi-disaster oriented and technology driven strategy through a culture of prevention, mitigation, preparedness and response. A disaster refers to a catastrophe, mishap, calamity or grave occurrence from natural or man-made causes, which is beyond the coping capacity of the affected community. DM involves a continuous and integrated process of planning, organising, coordinating and implementing measures which are necessary or expedient for:
 - Prevention of danger or threat of any disaster.
 - Mitigation or reduction of risk of any disaster or its severity or consequences.
 - Capacity building including research and knowledge management.
 - Preparedness to deal with any disaster.

A typical DM continuum comprises six elements; the pre-disaster phase includes prevention, mitigation and preparedness, while the postdisaster phase includes response, rehabilitation, reconstruction and recovery.

The objectives of the national policy on disaster management

- Promoting a culture of prevention, preparedness and resilience at all levels through knowledge, innovation and education.
- Encouraging mitigation measures based on technology, traditional wisdom and environmental sustainability.
- Mainstreaming disaster management into the developmental planning process.
- Establishing institutional and techno-legal frame works to create an enabling regulatory environment and a compliance regime.
 - Ensuring efficient mechanism for identification, assessment and monitoring of disaster risks.
- Developing contemporary forecasting and early warning systems backed by responsive and failsafe communication with information technology support.
- Promoting a productive partnership with the media to create awareness and contributing towards capacity development.
- Ensuring efficient response and relief with a caring approach towards the needs of the vulnerable sections of the society.
- Undertaking reconstruction as an opportunity to build disaster resilient structures and habitat for ensuring safer living.
- Promoting productive and proactive partnership with media in disaster management.

By guiding the country in managing and reducing disaster risks, the policy helps in several ways. It provides a clear framework for coordination among various stakeholders, including government agencies, NGOs, and the private sector, facilitating a unified response during emergencies. The emphasis on capacity building ensures that local communities are equipped to respond effectively to disasters, reducing vulnerability. Furthermore, the policy promotes research and the use of technology in disaster risk assessment and early warning systems, enhancing preparedness and response capabilities. Overall, the National Disaster Management Policy serves as a blueprint for building resilience and minimizing the impact of disasters in India.

- What are the advantages and drawbacks of involving stakeholders in disaster management?

The potential benefits of increased stakeholder participation include the following:

- ☐ Improved programme/project design
- ☐ Improved means of verifying the relevance and appropriateness of proposed interventions;

- ☐ Increased uptake of project services and greater willingness to share costs;
- ☐ Enhanced sustainability as a result of increased stakeholder ownership;
- ☐ Opportunity to foresee and/or resolve potential obstacles, constraints and conflicts;
- ☐ Opportunity to generate social learning and innovations based on field experience;
- ☐ Capacity-building of stakeholders and local institutions
- ☐ Improved means of ensuring that project benefits are distributed equitably;
- ☐ Strengthened working relations between stakeholders, government and civil society organizations and development partners.

The principal cost is the absence of stakeholder participation in programmes and projects. Lack of stakeholder participation can lead to:

- ☐ Higher up-front costs in terms of time and resources;
- ☐ Danger of undertaking poorly planned activities due to limited time, capacity, commitment or resources;
- ☐ Lack of political will on the part of governments to allow wide stakeholder participation because they fear loss of power or influence;
- ☐ Difficulty in reaching out to marginalized groups and ensuring

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that the true priorities and needs of poor and vulnerable groups are represented;

- ☐ Difficulty in identifying genuine representative non-governmental organizations (NGOs) and civil society organizations (CSOs);
- ☐ Creation of unrealistic expectations;
- ☐ Aggravating conflicts between stakeholder groups with different priorities/interests;
- ☐ Weak capacity of beneficiary and intermediary organisations.

- List the guiding principles of Sendai framework and explain.

GUIDING PRINCIPLES

The implementation of the present framework will be guided by the following principles, while taking into account national circumstances, and consistent with domestic laws as well as international obligations and commitments:

1. Each State has the primary responsibility to prevent and reduce disaster risk, including through international, regional, subregional, transboundary and bilateral cooperation. The reduction of disaster risk is a common concern for all States and the extent to which developing countries are able to effectively enhance and implement national disaster risk reduction policies and measures in the context of their respective circumstances and capabilities can be further enhanced through the provision of sustainable international cooperation;
2. Disaster risk reduction requires that responsibilities be shared by central Governments and relevant national authorities, sectors and stakeholders, as appropriate to their national circumstances and system of governance;

3. Managing the risk of disasters is aimed at protecting persons and their property, health, livelihoods and productive assets, as well as cultural and environmental assets, while promoting and protecting all human rights, including the right to development;
4. Disaster risk reduction requires an all-of-society engagement and partnership. It also requires empowerment and inclusive, accessible and non - discriminatory participation, paying special attention to people disproportionately affected by disasters, especially the poorest. A gender, age, disability and cultural perspective in all policies and practices; and the promotion of women and youth leadership; in this context, special attention should be paid to the improvement of organized voluntary work of citizens;
5. Disaster risk reduction and management depends on coordination mechanisms within and across sectors and with relevant stakeholders at all levels, and. it requires the full engagement of all State institutions of an executive and legislative nature at national and local levels and a clear articulation of responsibilities across public and private stakeholders, including business and academia, to ensure mutual outreach, partnership, complementarity in roles and accountability and follow-up;
6. While the enabling, guiding and coordinating role of national and federal State Governments remain essential, it is necessary to empower local authorities and local communities to reduce disaster risk, including through resources, incentives and decision-making responsibilities, as appropriate;
7. Disaster risk reduction requires a multi-hazard approach and inclusive risk-informed decision-making based on the open exchange and dissemination of disaggregated data, including by sex, age and disability, as well as on the easily accessible, up-to-date, comprehensible, science-based, non-sensitive risk information, complemented by traditional knowledge;
8. The development, strengthening and implementation of relevant policies, plans, practices and mechanisms need to aim at coherence, as appropriate, across sustainable development and growth, food security, health and safety, climate change and variability, environmental management and disaster risk reduction agendas. Disaster risk reduction is essential to achieve sustainable development;
9. While the drivers of disaster risk may be local, national, regional or global in scope, disaster risks have local and specific characteristics that must be understood for the determination of measures to reduce disaster risk;
10. Addressing underlying disaster risk factors through disaster riskinformed public and private investments are more cost-effective than primary reliance on post-disaster response and recovery, and contribute to sustainable development;
11. In the post-disaster recovery, rehabilitation and reconstruction phase it is critical to prevent the creation of and to reduce disaster risk by "Building Back Better" and increasing public education and awareness of disaster risk;
12. An effective and meaningful global partnership and the further strengthening of international cooperation, including the fulfilment of respective commitments of official development assistance by developed countries, are essential for effective disaster risk management;
13. Developing countries, in particular the least developed countries, small island developing States, landlocked developing countries and African countries, as well as middle-income and other countries facing specific disaster risk challenges need adequate, sustainable and timely provision of support, including through finance, technology transfer and capacity-building from developed countries and partners tailored to their needs and priorities, as identified by them.

- State the composition and role of National Executive Committee related with the National Policy on Disaster Management

The National Executive Committee

The National Executive Committee (NEC) comprises the Union Home Secretary as the Chairperson, and the Secretaries to the GOI in the Ministries/Departments of Agriculture, Atomic Energy, Defence, Drinking Water Supply, Environment and Forests, Finance (Expenditure), Health, Power, Rural Development, Science and Technology, Space, Telecommunications, Urban

Development, Water Resources and the Chief of the Integrated Defence Staff of the Chiefs of Staff Committee as members.

Secretaries in the Ministry of External Affairs, Earth Sciences, Human Resource Development, Mines, Shipping, Road Transport & Highways and Secretary, NDMA will be special invitees to the meetings of the NEC.

- The NEC is the executive committee of the NDMA, and is mandated to assist the NDMA in the discharge of its functions and also ensure compliance of the directions issued by the Central Government.
- NEC is to coordinate the response in the event of any threatening disaster situation or disaster.
- NEC will prepare the National Plan for Disaster Management based on the National Policy on Disaster Management.
- NEC will monitor the implementation of guidelines issued by NDMA.
- It will also perform such other functions as may be prescribed by the Central Government in consultation with the NDMA.